Project Planning & Management

DEPI – Junior Data Analytics Track

Project Title: US Accidents Data Analysis

Team Leader: Sama Wael Abdel Fatah Zaki

Team Members:

1. Sama Wael Abdel Fatah Zaki
2. Nour Mohamed Mahmoud Alhusseiny
3. Hanaa Khaled Abd Elhamid Ibrahim
4. Rehav Talaat Mohammed Alsalhy
5. Farha Khaled Ibrahim Abdelmotalb

Project Duration: 4 weeks

1/Project Plan

- Project Timeline

week	Planned Activities	Expected Outcome
week1	Data Collection, Dataset Understanding, and Project setup	Dataset Overview
week2	Data Cleaning, Preprocessing, and Handling Missing values	Cleaned & Processed Dataset
week3	Exploratory Data Analysis (EDA) and Trend/Pattern Analysis	EDA & Insights Report & Summary Tables
week4	Dashboard Building, Visualization, and Final Documentation	Interactive Dashboards, Final Project Report & Presentation

Gantt Chart – US Accidents Data Analysis

Project Task	Week 1	Week 2	Week 3	Week 4
Data Collection & Project Setup	■			
Dataset Understanding	■			
Data Cleaning & Preprocessing		■		
EDA & Trend Analysis			■	
Forecasting / Predictive Analysis			■	■
Dashboard Building & Visualization				■
Final Report & Presentation				■

- **Milestones**

Milestone	Description
Dataset Preparation Completed	The dataset has been collected and its structure has been understood.
Data Cleaning Completed	Missing values, duplicates, and data quality issues have been handled.
Data Analysis Completed	Exploratory Data Analysis (EDA) has been performed and key insights have been identified.
Visualization Completed	The project dashboards and visualizations have been prepared.
Project Submission Completed	The final project report and presentation have been prepared.

- **Deliverable**

Week	Deliverable	Description
Week 1	US Accidents Dataset (.csv)	The original dataset containing US accident records from 2016 to 2023.
Week 1	Dataset Overview	A summary of the dataset structure, columns, and key features.
Week 2	Cleaned & Processed Dataset	The dataset after handling missing values, duplicates, inconsistent data types, and other data quality issues.
Week 3	EDA & Insights Report	A report presenting the main findings, patterns, and trends identified during exploratory data analysis.
Week 3	Summary Tables & Charts	Tables and charts summarizing key results related to accident frequency, severity, weather conditions, time, and location.

Week 4	Accident Forecasting Results	Results of forecasting future accident trends and identifying potential high-risk periods using appropriate analytical techniques and available tools.
Week 4	Interactive Dashboards	Interactive visualizations presenting project findings using Power BI and Tableau.
Week 4	Final Project Report & Presentation	The final report and presentation summarizing the project objectives, methodology, analysis, key findings, and recommendation

- **Resource allocation**

Resource	**Purpose**
US Accidents Dataset (CSV)	**Provides historical accident data for analyzing accident patterns, trends, and high-risk locations.**
Microsoft Excel	**Used for initial data inspection, data organization, preliminary analysis, and forecasting future accident trends.**
Power BI	**Used to create interactive dashboards, visualize accident patterns, and identify geographic accident hotspots.**
Tableau	**Used to create visualizations and present analytical insights.**
Power Query	**Used to clean, transform, and preprocess the dataset, if applicable.**
Team Members	**Collaborate on data preparation, analysis, visualization, documentation, and project delivery.**
Four-Week Schedule	**Organizes project activities and helps the team meet deadlines.**

2/Task Assignment &Roles

Team Member	Assigned Tasks	Tools
Sama (Team Leader)	Data Cleaning, Forecasting, Data modeling, Team Coordination	Excel, Tableau
Nour	Data Collection, Dataset Documentation, Data Modeling, Dashboard Development	Power BI , Tableau
Hanaa	Data Cleaning, EDA, Insights, Dashboard Development	Power BI, Tableau
Rehav	Data Cleaning, EDA, Trend Analysis, Insights	Excel , Tableau
Farha	Data Visualization, Final Documentation	Tableau

3/Risk Assessment &Mitigation Plan

Potential Risk	**Mitigation Strategy**
Missing Values and Data Quality Issues	**Use Power Query or Excel to handle missing values, duplicates, and inconsistent data types.**
Large Dataset Size	**Remove unnecessary columns and apply appropriate filters to improve processing performance.**
Incorrect Date and Time Formats	**Convert date and time columns to the correct data types before performing temporal analysis.**
Incorrect Relationships Between Tables	**Review table keys and verify relationships to ensure accurate analysis and filtering.**

Forecasting Inaccuracy	**Use appropriate forecasting techniques, review historical trends, and compare predicted results with actual data when available.**
Difficulty Identifying High-Risk Locations	**Analyze accident frequency and severity by location and use geographic visualizations to identify potential hotspots.**
Time Constraints	**Follow the project timeline, distribute tasks clearly, and monitor progress regularly.**

4/KPIs (Key Performance Indicators)

KPI	Description
Total Accidents Count	**Measures the total number of recorded traffic accidents within the selected period and geographic area.**
Average Accident Severity	**Measures the average severity level of recorded accidents, ranging from 1 to 4.**
Accidents by Weather Condition	**Shows the number of accidents under different weather conditions.**
Accidents by Time of Day	**Shows how accidents are distributed across different hours and time periods.**
Average Accident Duration	**Measures the average time between the recorded accident start and end times, based on valid timestamps.**
Accidents at Intersections and Traffic Signals	**Measures the number or percentage of accidents associated with intersections and traffic signals.**
Accidents by Location	**Shows the distribution of accidents across states, cities, or geographic areas to help identify potential accident hotspots.**
Forecasted Accident Count	**Estimates the expected number of accidents in future periods based on historical accident trends.**